Acids, bases and salts

30 questions: 15 multiple choice, 15 structured.

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1 Core The characteristic properties of acids and bases [1]

A student adds a few drops of three different indicators to separate samples of dilute nitric acid.

Which row shows the colour of each indicator in the dilute nitric acid?

	litmus	**thymolphthalein**	**methyl orange**
A	red	colourless	red
B	blue	blue	yellow
C	red	blue	red
D	red	colourless	yellow

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2 Core The characteristic properties of acids and bases [4]

Dilute sulfuric acid and aqueous sodium hydroxide are both used in a school laboratory.

(a) Give the formula of the ion that is present in all aqueous solutions of acids. [1]

(b) Give the formula of the ion that is present in all aqueous solutions of alkalis. [1]

(c) Copper(II) oxide is a base, but it is not an alkali. Explain why. [2]

3 Core The characteristic properties of acids and bases [1]

Four solutions, W, X, Y and Z, are tested with universal indicator paper. The pH values are shown.

solution	W	X	Y	Z
pH	12	6	2	8

Which list shows the solutions in order of **increasing** hydrogen ion concentration?

A Y, X, Z, W

B W, Z, X, Y

C W, X, Z, Y

D Z, X, W, Y

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4 Extended The characteristic properties of acids and bases [1]

Hydrogen chloride gas reacts with ammonia gas to form a white solid.

$HCl(g) + NH_3(g) \rightarrow NH_4Cl(s)$

Which statement about this reaction is correct?

A HCl acts as a base because it accepts a proton.

B NH_3 acts as an acid because it donates a proton.

C NH_3 acts as a base because it accepts a proton.

D HCl acts as an acid because it accepts a proton.

5 Core Oxides [1]

Element E is a shiny solid. It conducts electricity and is malleable.

Which row describes element E and the type of oxide it is most likely to form?

	type of element	**type of oxide**
A	metal	basic
B	metal	acidic
C	non-metal	basic
D	non-metal	acidic

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6 Core The characteristic properties of acids and bases [7]

A student adds a piece of magnesium ribbon to some dilute hydrochloric acid in a test-tube.

(a) Name the gas produced and the salt formed in this reaction. [2]

(b) Write the balanced symbol equation for the reaction between magnesium and hydrochloric acid. [2]

(c) In a second experiment, the student adds zinc carbonate powder to dilute hydrochloric acid.

Complete the word equation for this reaction.

zinc carbonate + hydrochloric acid → ..... + ... + ... [2]

(d) State **one** observation the student would make during the reaction in (c). [1]

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7 **Extended** The characteristic properties of acids and bases [7]

Hydrochloric acid is a strong acid. Ethanoic acid is a weak acid.

(a) Define the terms *strong acid* and *weak acid*. [2]

(b) Write the equation for the dissociation of ethanoic acid in water. Include state symbols. [2]

(c) A student has a solution of hydrochloric acid and a solution of ethanoic acid. Both solutions have the same concentration. Using universal indicator paper, the student finds that one solution has pH 1 and the other has pH 3.

Identify which acid has pH 3. Explain your answer. [3]

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8 Core The characteristic properties of acids and bases [6]

Ammonium sulfate is used as a fertiliser. Calcium hydroxide (slaked lime) is a base that gardeners sometimes add to soil.

When ammonium sulfate is warmed with calcium hydroxide, a gas is given off.

(a) Name the gas given off. [1]

(b) The gas is alkaline. State the colour change seen when damp red litmus paper is held in the gas. [1]

(c) Complete the word equation for the reaction.

ammonium sulfate + calcium hydroxide → ..... + ... + water [2]

(d) Suggest why a gardener should **not** spread calcium hydroxide on the soil at the same time as ammonium sulfate fertiliser. [2]

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9 Core Preparation of salts [7]

A student prepares crystals of copper(II) sulfate. The student adds copper(II) oxide, a little at a time, to warm dilute sulfuric acid and stirs. The student stops adding copper(II) oxide when some black solid remains unreacted.

(a) Write the balanced symbol equation for the reaction between copper(II) oxide and sulfuric acid. [1]

(b) Explain why the student adds copper(II) oxide until some remains unreacted. [2]

(c) Name the method used to remove the unreacted copper(II) oxide from the mixture. [1]

(d) Describe how the student can obtain pure, dry crystals of copper(II) sulfate from the solution left after (c). [3]

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10 Core Preparation of salts [1]

Which pair of salts are **both** insoluble in water?

A calcium chloride and lead(II) nitrate

B sodium carbonate and barium sulfate

C copper(II) carbonate and potassium sulfate

D barium sulfate and silver chloride

11 Core The characteristic properties of acids and bases [1]

A few drops of thymolphthalein are added to a colourless solution. The solution turns blue.

Which row shows the colour of litmus and of methyl orange in separate samples of the same solution?

	litmus	**methyl orange**
A	blue	red
B	red	yellow
C	blue	yellow
D	red	red

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12 Core The characteristic properties of acids and bases [6]

Indigestion is caused by too much hydrochloric acid in the stomach. Some indigestion tablets contain magnesium hydroxide, which is a base.

(a) Name the salt formed when magnesium hydroxide reacts with hydrochloric acid. [1]

(b) Write the balanced symbol equation for the reaction between magnesium hydroxide and hydrochloric acid. [2]

(c) Other indigestion tablets contain calcium carbonate instead of magnesium hydroxide.

Name the gas produced when calcium carbonate reacts with hydrochloric acid. Suggest one reason why producing this gas could be a disadvantage for a person taking the tablets. [2]

(d) State the name of the type of reaction that happens between an acid and a base. [1]

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13 Core The characteristic properties of acids and bases [1]

Dilute nitric acid is added to aqueous potassium hydroxide until the solution is exactly neutral.

Which ions are present in the neutral solution in the largest amounts?

A K^+ and NO_3^-

B H^+ and OH^-

C H^+ and NO_3^-

D K^+ and OH^-

14 Extended The characteristic properties of acids and bases [1]

P and Q are two different acids. Aqueous solutions of P and Q are made with the same concentration.

The solution of P has pH 1. The solution of Q has pH 4.

Which statement is correct?

A P is partially dissociated in aqueous solution.

B The solution of Q has a higher concentration of H^+ ions than the solution of P.

C Q is partially dissociated in aqueous solution.

D P and Q are both strong acids because both solutions have a pH below 7.

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15 Extended Oxides [1]

Oxide X reacts with dilute hydrochloric acid. Oxide X also reacts with aqueous sodium hydroxide. A salt and water are formed in both reactions.

What is oxide X?

A calcium oxide

B carbon dioxide

C zinc oxide

D copper(II) oxide

16 **Extended** The characteristic properties of acids and bases [6]

(a) Define an *acid* and a *base* in terms of protons. [2]

(b) The dissociation of hydrochloric acid is shown by the equation.

$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$

State what the single, one-way arrow shows about hydrochloric acid. [1]

(c) When hydrochloric acid neutralises aqueous sodium hydroxide, the reaction is

$H^+(aq) + OH^-(aq) \rightarrow H_2O(l)$

Identify the species that acts as a base in this reaction. Explain your answer. [2]

(d) Ethanoic acid dissociates in water.

$CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$

In the reverse reaction, H^+ ions and CH_3COO^- ions join to form CH_3COOH.

Identify the species that acts as a base in the reverse reaction. [1]

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17 Core Oxides [6]

A student is given four oxides: calcium oxide, CaO, copper(II) oxide, CuO, carbon dioxide, CO_2, and sulfur dioxide, SO_2.

(a) Classify each of the four oxides as acidic or basic. [2]

(b) State how the type of oxide an element forms is related to whether the element is a metal or a non-metal. [2]

(c) Copper(II) oxide does not dissolve in water, so it has no effect on the pH of water.

Suggest how the student could show by an experiment that copper(II) oxide is a basic oxide. Include the result the student would expect. [2]

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18 **Extended** Preparation of salts [7]

Barium sulfate is an insoluble salt. It is used in hospitals to help X-ray images of the stomach show up clearly. It can be made in the laboratory by precipitation.

(a) Name **two** aqueous solutions that could be mixed to make a precipitate of barium sulfate. [2]

(b) Write the ionic equation for the formation of barium sulfate. Include state symbols. [2]

(c) Describe how a pure, dry sample of barium sulfate could be obtained from the mixture after precipitation. [3]

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19 Core Preparation of salts [1]

Which statement defines an *anhydrous* substance?

A a substance that contains no water

B a substance that is chemically combined with water

C a substance that has been dissolved in water

D a substance that absorbs water vapour from the air

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20 Core The characteristic properties of acids and bases [5]

A student tests five household solutions with universal indicator paper. The results are shown.

solution	pH
drain cleaner	13
vinegar	3
salt solution	7
baking soda solution	9
rainwater	5

(a) Identify the solution that is neutral. [1]

(b) Identify the solution with the highest concentration of hydroxide ions. [1]

(c) Put vinegar, rainwater and salt solution in order of **increasing** hydrogen ion concentration. [1]

(d) Describe how the student used universal indicator paper to find the pH of one of the solutions. [2]

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21 Core The characteristic properties of acids and bases [1]

Which substance reacts with dilute sulfuric acid to form a salt, water **and** a gas?

A copper

B sodium carbonate

C magnesium

D zinc oxide

22 Extended The characteristic properties of acids and bases [1]

Which equation shows how ethanoic acid behaves as a weak acid in water?

A $CH_3COOH(aq) \rightarrow H^+(aq) + CH_3COO^-(aq)$

B $CH_3COOH(aq) \rightleftharpoons CH_3COO^+(aq) + H^-(aq)$

C $CH_3COOH(aq) \rightleftharpoons CH_3CO^+(aq) + OH^-(aq)$

D $CH_3COOH(aq) \rightleftharpoons H^+(aq) + CH_3COO^-(aq)$

23 Core Preparation of salts [1]

A student wants to make pure crystals of potassium nitrate.

Which method is the most suitable?

A Add excess potassium metal to dilute nitric acid, filter, then crystallise the filtrate.

B Add excess aqueous potassium hydroxide to dilute nitric acid, filter, then crystallise the filtrate.

C Add excess solid potassium carbonate to dilute hydrochloric acid, filter, then crystallise the filtrate.

D Find the volume of dilute nitric acid needed to neutralise aqueous potassium hydroxide by titration with an indicator, repeat using the same volumes without the indicator, then crystallise the solution.

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24 Extended Preparation of salts [1]

Blue crystals of hydrated copper(II) sulfate have the formula $CuSO_4 \cdot 5H_2O$.

Which statement about these crystals is correct?

A The crystals are a mixture of copper(II) sulfate and liquid water, and the water can be removed by filtration.

B The $5H_2O$ represents water of crystallisation: water molecules present in the hydrated crystals.

C The crystals contain no water once they have been dried with filter paper.

D The crystals are anhydrous because the water is chemically combined.

25 Core The characteristic properties of acids and bases [1]

Which substance is an alkali?

A copper(II) oxide

B potassium hydroxide

C iron(III) hydroxide

D magnesium carbonate

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26 Core The characteristic properties of acids and bases [6]

A student puts 25.0 cm^3 of aqueous sodium hydroxide in a conical flask and adds a few drops of thymolphthalein. Dilute hydrochloric acid is then added from a burette until the indicator changes colour.

(a) State the colour of the thymolphthalein in the flask at the start. [1]

(b) State the colour of the thymolphthalein when a slight excess of hydrochloric acid has been added. [1]

(c) Write the ionic equation for the neutralisation reaction. Include state symbols. [2]

(d) Name the salt formed, and state the pH of the solution in the flask when the sodium hydroxide has been exactly neutralised. [2]

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27 **Extended** Oxides [7]

A student has two white powders, P and Q. One is aluminium oxide and the other is magnesium oxide. Each powder is warmed with dilute hydrochloric acid, and a fresh sample of each is warmed with aqueous sodium hydroxide. The results are shown.

powder	warmed with dilute hydrochloric acid	warmed with aqueous sodium hydroxide
P	reacts; a colourless solution forms	reacts; a colourless solution forms
Q	reacts; a colourless solution forms	no reaction

(a) Identify which powder is aluminium oxide. Explain your answer. [2]

(b) Describe what is meant by an *amphoteric oxide*. [2]

(c) Suggest the name of another oxide that would give the same results as P. [1]

(d) Classify powder Q as acidic, basic or amphoteric. Explain your answer in terms of the element it contains. [2]

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28 Core The characteristic properties of acids and bases [7]

Three bottles contain colourless solutions labelled J, K and L. One is dilute hydrochloric acid, one is aqueous sodium hydroxide and one is aqueous sodium chloride. A student does three tests on each solution. The results are shown.

test	**J**	**K**	**L**
add a piece of magnesium ribbon	bubbles of gas	no change	no change
add a few drops of thymolphthalein	colourless	blue	colourless
warm with solid ammonium chloride	no gas	gas given off that turns damp red litmus paper blue	no gas

(a) Identify solutions J, K and L. [3]

(b) Name the gas given off in the magnesium test with J, and the gas given off when K is warmed with ammonium chloride. [2]

(c) Solution K is added to a sample of J until the mixture is exactly neutral. A piece of magnesium ribbon is then added to this neutral mixture.

Predict what would be seen. Explain your answer. [2]

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29 **Extended** Preparation of salts [6]

Epsom salts are crystals of hydrated magnesium sulfate, $MgSO_4 \cdot 7H_2O$. A student heats some Epsom salts strongly in an evaporating basin. The student weighs the basin and contents several times and stops heating when the mass no longer decreases. A white powder remains.

(a) Define the term *hydrated substance*. [1]

(b) State the name given to the water molecules shown as $7H_2O$ in the formula $MgSO_4 \cdot 7H_2O$. [1]

(c) Give the formula of the white powder that remains. State whether it is hydrated or anhydrous. [2]

(d) Explain why the mass decreases during heating, and why the student heats until the mass no longer decreases. [2]

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30 **Extended** Preparation of salts [8]

A student wants to make three sulfate salts: lead(II) sulfate, potassium sulfate and zinc sulfate.

The student can choose from these three methods.

- method 1: titration of an acid with an alkali
- method 2: adding an excess of an insoluble solid (a metal, an insoluble base or an insoluble carbonate) to an acid, then filtering
- method 3: precipitation by mixing two solutions

(a) State the most suitable method (1, 2 or 3) for making each salt. [3]

(b) Name **two** aqueous solutions that could be mixed to make lead(II) sulfate. [2]

(c) Name **one** solid that could be added in excess to dilute sulfuric acid to make zinc sulfate. [1]

(d) Explain why potassium sulfate is made by titration rather than by adding an excess of a solid potassium compound to the acid. [2]

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